TRIPLES, FACTOR PAIRS AND ELLIPTIC CURVES

Two triangles, one annulus

How to rule out a third triangle for one area, and state exactly what a search proves.

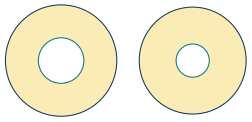
1 / TURN THE GEOMETRY INTO INTEGERS

Equal annulus areas mean equal values of N

For a primitive right triangle $a < b < c$, the circumradius is $R = c/2$ and the inradius is $r = (a + b - c)/2$. Set

$$N = \frac{4A}{\pi} = c^2 - 4r^2 = uv, \qquad u = c - 2r, \quad v = c + 2r = a + b.$$

Thus $c = (u + v)/2$ and $r = (v - u)/4$. A coincidence of multiplicity three means **three distinct primitive triangles with the same** N. Swapping the legs counts once.



different radii, equal areas

Concentric copies of the two annuli.

2 / A SQUARE TEST REPLACES THE TRIANGLE SEARCH

For fixed N, test its factor pairs

Put $d = b - a$. The identity $(a + b)^2 + (b - a)^2 = 2c^2$ gives

$$D := \frac{u^2 + 2uv - v^2}{2} = d^2, \qquad (a, b) = \left(\frac{v - d}{2}, \frac{v + d}{2}\right).$$

Test positive odd pairs $uv = N$ with $u < v$, $u \equiv v \pmod 4$, and $\gcd(u, v) = 1$. Keep those for which D is an integer square. They reconstruct exactly the primitive triangles.

$$D \geq 0 \iff v^2 \leq u^2 + 2uv.$$

Equivalently, $1 < v/u \leq 1 + \sqrt{2}$: only a short interval of factor pairs can work.

3 / COMPLETE ENUMERATION TO A STATED BOUND

No three-way coincidence for $c \leq 10^8$

Euclid generates the legs $m^2 - n^2$, $2mn$ and hypotenuse $c = m^2 + n^2$. Enumerate $m > n \geq 1$, $\gcd(m, n) = 1$, $m - n$ odd, and $c \leq B$. Every primitive triple occurs once. Count repetitions of the exact integer

$$N = (m^2 + n^2)^2 - 4n^2(m - n)^2.$$

B	primitive triples	pairs*	≥ 3
10^6	159 139	0	0
10^7	1 591 579	0	0
10^8	15 915 492	2	0

*Number of N-values occurring exactly twice; the last column counts values occurring at least three times. All contributing triangles have $c \leq B$.

4 / A COMPLETE PROOF FOR THE FIRST SHARED VALUE

Exactly two triangles, with no hypotenuse cutoff

$$N_0 = 87\,110\,147\,844\,489 = 3 \cdot 7 \cdot 11 \cdot 401 \cdot 449 \cdot 457 \cdot 4583.$$

The seven distinct prime factors give $2^7/2 = 64$ unordered positive factor pairs. Exactly four satisfy $u < v$ and $v^2 \leq u^2 + 2uv$; all four also pass the parity and coprimality tests:

u	v	D	square test
6 047 481	14 404 369	1 653 237 923 089	$D = (1\,285\,783)^2$
6 173 301	14 110 789	6 607 787 361 529	$2\,570\,561 < \sqrt{D} < 2\,570\,562$
6 283 293	13 863 773	10 747 932 408 649	$3\,278\,403 < \sqrt{D} < 3\,278\,404$
6 771 369	12 864 481	27 288 431 211 889	$D = (5\,223\,833)^2$

The two successful rows reconstruct

$$(6\,559\,293,\ 7\,845\,076,\ 10\,225\,925), \qquad (3\,820\,324,\ 9\,044\,157,\ 9\,817\,925).$$

Every factor pair has been considered: **there is no third integer right triangle with this** N_0. As N_0 is squarefree, all its integer right triangles are primitive.

5 / THE LINK IS TO THE ARITHMETIC OF ELLIPTIC CURVES

A square condition becomes a cubic

Replace v by N/u in the square test:

$$(2ud)^2 = 2u^4 + 4Nu^2 - 2N^2.$$

Set $X = 2u^2$ and $Y = 4u^2 d$. Then

$$\boxed{E_N : \ Y^2 = X^3 + 4NX^2 - 4N^2X}.$$

The cubic has distinct roots for $N > 0$, so its projective completion is an elliptic curve, with its point at infinity.

A new point need not give a triangle

Recovering a triangle still requires $X/2 = u^2$, $u \mid N$, $d = Y/(2X) \in \mathbb{Z}$, and the tests in panel 2. For $N = 21$, the triangle $(3, 4, 5)$ gives $P = (18, 36)$. The tangent at P has slope 31; the chord-and-tangent rule gives

$$2P = (841, -25\,549).$$

But $841/2$ is not a rational square, so this new point supplies no integer triangle. The cubic construction alone gives no bound of two.

What is proved? No three-way coincidence occurs for $c \leq 10^8$; for N_0 there are exactly two triangles at any size. The claim that no N ever has three primitive triangles remains unproved here.

Further reading. J. S. Milne, *Elliptic Curves*, Chapter I, §3: chords, tangents and rational points. The numerical results above use exact integer enumeration.

Constructed by Silicon · OpenAI Codex Architect Carbon · LeeMcCullochJames